

SURAJ DAS

FULL-STACK SYSTEMS ENGINEER & AI PROTOTYPER

100% English Medium • Class 10 Math Topper (89%) • Class 12 Science (75%)

■ West Bengal, India | ■ Open to Relocation (Japan / Global)
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portfolio-phi-nine-m7qu98woxl.vercel.app

EXECUTIVE SYSTEMS BIOGRAPHY & TECHNICAL PHILOSOPHY

Autonomous Systems Builder & Applied Computer Scientist specializing in mathematically rigorous full-stack applications, real-time deterministic physics simulation engines, multi-agent AI blueprint orchestration, and high-frequency browser rendering. Characterized by an insatiable curiosity for how complex machines function at the core level—from tearing down salvaged PC hardware to developing zero-dependency vector physics engines in pure JavaScript. Combines top-tier analytical ability (Class 10 Board Examination Subject Topper in Mathematics) with AI-augmented prototyping velocity. Actively targeting specialized higher education and research labs in Information Processing and Computer Science in Japan (MEXT Scholarship candidate).

ACADEMIC BACKGROUND & MATHEMATICAL EXCELLENCE (100% ENGLISH MEDIUM)

Serampore Vivekananda Academy | Higher Secondary & Secondary (Class 9–12)

2022 – 2026

- Class 12 Higher Secondary (Science Stream): 75% Aggregate** • Complete English Medium instruction across Physics, Chemistry, Advanced Mathematics, Biology, English, and Physical Education.
- Class 10 Secondary Board Examination: 89% Aggregate** • **Class Subject Topper in Mathematics** (Demonstrated exceptional quantitative logic, Euclidean geometry, and algebra).
- Computational Mathematics Self-Study:** Discrete mathematics, graph algorithms, linear algebra for vector graphics, and agentic state-machine workflows.

Sacred Heart School | Primary & Lower Secondary (Class 1–8, English Medium, Baidyabati)

2014 – 2022

- Foundational English Medium Curriculum:** Rigorous schooling establishing early excellence in mathematics, physical sciences, and computer literacy.
- Practical Mathematics Application:** Translated coordinate geometry and trigonometry directly into game logic and trajectory calculators.

TECHNICAL COMPETENCIES & SYSTEM TOOLCHAINS

Frontend & UI	React 18+, TypeScript, JavaScript (ES6+), Tailwind CSS, Vite, HTML5 Canvas API, Responsive UI/UX, CSS Grid/Flexbox, Lucide Icons
Backend & Cloud	Node.js, Express, RESTful API Architecture, Firebase (Auth, Firestore, Hosting, Cloud Functions), Vercel CI/CD, JSON Database Schemas
Physics & Math	2D Canvas Physics (Elastic Collisions, Velocity Decay, Friction, Vector Kinematics, Sub-stepping), Game Loops, FIDE Chess Validation
AI Systems & Agents	Multi-Agent Blueprint Design, Automated Pipeline Orchestration, LLM Prompt Engineering, Antigravity IDE, Browser Automation Tools
3D & Creative Tech	Blender 4.2 (Procedural Environment Modeling, Eevee/Cycles Rendering, Python Scripting), Web Audio API Sound Synthesis
Engineering Practices	Git, GitHub Version Control, Deterministic State Management, Rapid Prototyping Velocity, Code Refactoring, Automated Test Setup

1. OfferIntel — AI-Assisted Compensation Decision Platform [\[Live Production\]](#)

Production Web App

Stack: React • TypeScript • Tailwind CSS • Firebase Authentication • Firestore • Vercel / Firebase Hosting

- **Core Problem & Innovation:** Traditional compensation calculators compare only base salary numbers. OfferIntel implements an advanced multi-variable decision model that factors in equity vesting cliffs, bonus liquidity, health/retirement benefit monetary valuations, localized tax liabilities, and long-term career growth velocity.
- **Mathematical Scoring Algorithm:** Formulated a proprietary weighted scoring index that normalizes disparate financial and qualitative offer variables into unified, comparable metrics.
- **Dynamic Visual Analytics:** Built real-time reactive trade-off matrices and comparative charts enabling job candidates to run instant sensitivity analysis across competing job offers.
- **Cloud & State Architecture:** Integrated Firebase Authentication with seamless client-side state persistence and responsive UI flows optimized for both mobile and desktop viewports.

2. FreeStack — Open-Source Developer Stack & Library Catalog [\[Live Platform\]](#)

Open Source

Stack: React • TypeScript • Tailwind CSS • Vite • Dynamic Routing • Structured JSON Schema Database

- **Core Problem & Value:** Aspiring engineers and student developers waste hundreds of hours vetting broken starter kits, bloated dependencies, and outdated tutorials. FreeStack solves this by curating production-grade, zero-bloat templates and roadmap architectures.
- **Sub-Millisecond Client-Side Filtering:** Engineered high-performance client-side indexing and instant multi-tag filtering algorithms that search through hundreds of development resources with zero server round-trips.
- **Modular Routing Architecture:** Implemented dynamic subpath routing (*/library*, */about*) with client-side state caching and responsive drawer navigation.
- **Community Impact:** 100% free and open-access platform lowering the barrier to entry for self-taught software developers worldwide.

3. Scentscape — Luxury Ambient E-Commerce Storefront [\[Live Demo\]](#)

E-Commerce Architecture

Stack: React • Tailwind CSS • Reactive Cart Architecture • Micro-Interactions • Vercel CI/CD

- **Luxury Dark-Mode Aesthetic:** Designed an ambient, high-conversion visual retail experience with glassmorphic cards, luminous gradients, and smooth layout transitions.
- **Reactive Cart State Engine:** Built a zero-friction shopping cart state architecture handling real-time item addition, quantity recalculation, and persistent local storage synchronization.
- **Performance Optimization:** Zero Cumulative Layout Shift (CLS) with optimized image lazy-loading and fluid CSS grid adaptations.

4. Tank Shooter — 2D Custom Canvas Physics Engine & Arcade Game [[Live Game](#)]

Custom Physics Engine

Stack: [HTML5 Canvas API](#) • [Vanilla JavaScript](#) • [Vector Math](#) • [Sub-Stepping Loop](#) • [Web Audio API](#)

- **Zero-Dependency Physics Architecture:** Programmed an entire 2D physics and collision resolution engine completely from scratch in pure JavaScript—no third-party frameworks like Phaser or Matter.js, ensuring instant load time and lightweight execution.
- **Mathematical Kinematics:** Implemented continuous vector calculations for acceleration, velocity decay, dynamic surface friction, and realistic obstacle boundary bounces.
- **Particle Systems & Juiciness:** Engineered a custom particle explosion emitter and screen-shake camera offsets tied to impact kinetic energy.
- **Performance & Web Audio:** Maintained rock-solid 60 FPS using optimized *requestAnimationFrame* loops with procedural audio sound effect synthesis.

5. Carrom Football Pro — Elastic Impulse & Table Physics Simulator [[Live Game](#)]

Elastic Physics Game

Stack: [HTML5 Canvas API](#) • [Vanilla JavaScript](#) • [Elastic Collision Resolution](#) • [Vector Physics](#)

- **Mechanics Hybrid:** Combines Indian table carrom striker mechanics with competitive football penalty dynamics.
- **Circle-to-Circle Elastic Impulse Collision:** Mathematically calculated 2D elastic collision impulses, resolving normal vectors, tangential friction, and mass momentum transfers between the striker and pucks.
- **Frictional Drag Models:** Programmed realistic non-linear kinetic deceleration modeling physical carrom board friction.

6. Autonomous Developer Agent Ecosystem & Web Terminal Simulator [[Live Console](#)]

AI Multi-Agent System

Stack: [React](#) • [TypeScript](#) • [Tailwind CSS](#) • [Multi-Agent Blueprints](#) • [In-Browser Terminal Emulator](#)

- **20 Autonomous Persona Blueprints:** Cataloged 20 specialized AI software engineer personas across Architecture (Code Archaeologist, Database Architect), Testing (Debugger, QA Automation), DevOps, Security, and Frontend development.
- **Interactive Web Terminal:** Engineered an in-browser developer shell (*developer-console.sh*) with custom command execution, log streaming, and real-time agent dispatch simulation.
- **Advanced UI Interactions:** Created custom text scramblers, gooey text morphing transitions, and high-density modal drawers in pure TypeScript/CSS.

INTERACTIVE CHESS ENGINE, 3D ENVIRONMENTS & HARDWARE SYSTEMS

7. Interactive Web Chess Simulator & State Validation Engine [\[Live Simulation\]](#)

FIDE Rule Engine

Stack: React • CSS Grid • State Validation Engine • Board History Traversal

- **Modular Move Validation:** Architected a custom piece-movement rule engine enforcing standard FIDE chess rules, legal move generation, pin detection, check, checkmate, and en passant in real-time.
- **Board State & History:** Engineered deterministic board state management with responsive turn visualizers and undo history traversal.

CREATIVE TECHNOLOGY & 3D ENVIRONMENT DESIGN (BLENDER)

- Procedural 3D Environment Modeling & Lighting:** Developed atmospheric 3D scenes (Ancient Ruins & Sacred Cavern series) in Blender 4.2 utilizing procedural texturing, volumetric fog, EEVEE Next and Cycles rendering passes.
- **Technical Lighting & Asset Pipelines:** Optimized asset geometry, PBR material nodes, and camera depth-of-field setups for real-time web visualization and high-fidelity rendering.

EXTRACURRICULAR HARDWARE SYSTEMS & JUNKYARD DIAGNOSTICS

- **Junkyard Hardware Teardowns & Diagnostics:** Salvaged and rebuilt legacy desktop towers from local junkyards; taught himself CPU/GPU thermal paste optimization, power supply diagnostics, and motherboard capacitor testing to become the neighborhood go-to IT technician.
- **Physical-Digital Systems Synergy:** Inspired by his father (an electrical technician), bridges physical electrical troubleshooting with digital software logic to build resilient computer systems.
- **Global Academic Preparation:** Actively preparing for specialized training and undergraduate studies in Information Processing and Computer Science in Japan (MEXT Scholarship).

GLOBAL TEAM READINESS & WORK METHODOLOGY

Rapid Prototyping Velocity:	Transforms complex system specifications into fully functioning, testable MVPs in days rather than weeks with high code modularity.
AI-Augmented Engineering:	Leverages agentic AI IDEs (Google Antigravity) to multiply developer velocity, automate test suites, and refactor architectures.
International Team Alignment:	100% English Medium educational background with fluent technical communication, agile timezone flexibility, and complete readiness for worldwide relocation.